

Introduction to (Statistical) Machine Learning

Problem Set Three

Summer 2026, IIM B

Due by 11:59 pm on 10 July 2026

Instructions. All four problems are required. For theoretical problems, using LLM's is not allowed. You are free to use them for the coding problems.

1. This problem studies the bootstrap in two simple examples.

- a) Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} N(\mu, 1)$. Let

$$\theta = e^\mu$$

and let

$$\hat{\theta} = e^{\bar{X}}$$

be the maximum likelihood estimator of θ .

Generate a dataset of size $n = 100$ from $N(5, 1)$. Using this dataset, use the nonparametric bootstrap to estimate the standard error of $\hat{\theta}$.

- b) Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Uniform}(0, \theta)$. The maximum likelihood estimator of θ is

$$\hat{\theta} = X_{\max} = \max\{X_1, \dots, X_n\}.$$

Generate a dataset of size $n = 50$ from $\text{Uniform}(0, 1)$.

Find the exact distribution of $\hat{\theta}$ when $\theta = 1$. Compare the true distribution of $\hat{\theta}$ to the histogram of $\hat{\theta}$ obtained from repeated simulations.

Now use the bootstrap on your generated dataset to estimate the standard error of $\hat{\theta}$. Compare the true standard error of $\hat{\theta}$ to the bootstrap standard error. What do you observe? Explain why this happens.

2. The kidney cancer dataset is analyzed in Section 2.7 of Gelman et al. [2]. In that section, the authors use Poisson likelihoods with Gamma priors, whereas in class we worked with Binomial likelihoods and Beta priors. Here, we revisit the in-class analysis, replacing the Binomial–Beta model with a Poisson–Gamma model.

As in class, let X_i denote the total death count due to kidney cancer for county i in the period 1980–89. Also let n_i be the average population for county i in the same time period. Let θ_i be an unknown parameter representing the death rate for county i in units of deaths per person per decade.

- a) Argue that

$$X_i \sim \text{Poisson}(n_i \theta_i).$$

You can use the fact that $\text{Bin}(n, p)$ is well-approximated by $\text{Poisson}(np)$ provided np^2 is small.

- b) Consider the model

$$\theta_i \stackrel{\text{i.i.d}}{\sim} \text{Gamma}(\alpha, \beta), \quad X_i | \theta_i \sim \text{Poisson}(n_i \theta_i).$$

Treat α and β as unknown parameters, and estimate them by maximizing the marginal likelihood. Use a numerical grid-based discretization approach to compute the maximum marginal likelihood estimates $\hat{\alpha}$ and $\hat{\beta}$.

- c) Use $\hat{\alpha}$ and $\hat{\beta}$ from part (b) to compute the posterior mean estimates of θ_i , $i = 1, \dots, N$. Compare these estimates with those derived in class using the Beta-Binomial model. Identify the top 100 counties in terms of $\hat{\theta}_i$, and compare this list with that derived in class using the Beta-Binomial model.
- d) Consider the full Bayes model

$$\begin{aligned} \alpha, \beta &\sim \text{uninformative priors}, \\ \theta_i &\stackrel{\text{i.i.d}}{\sim} \text{Gamma}(\alpha, \beta), \\ X_i | \theta_i &\sim \text{Poisson}(n_i \theta_i). \end{aligned} \tag{1}$$

What uninformative priors would you use for α and β , and why? Without using any probabilistic programming library, implement Bayesian inference in this model and compute posterior mean estimates of $\theta_1, \dots, \theta_N$. You can use a discretization-based approach similar to the one discussed in class for the Beta-Binomial model.

- e) Use PyMC or another probabilistic programming library to obtain posterior samples for the model (1). Use these samples to compute posterior mean estimates of $\theta_1, \dots, \theta_N$. Compare the posterior means with those obtained in part (d).
3. Consider the following baseball dataset from Efron [1, Chapter 1]. The goal is to use the players' early season performance, indicated by the second column, to predict their end-of-season performance, indicated by the third column.

Use the following model for this dataset:

$$\log a, \log b \stackrel{\text{i.i.d}}{\sim} \text{uniform}(-\infty, \infty),$$

$$\theta_i | a, b \sim \text{Beta}(a, b),$$

and

$$H_i \stackrel{\text{ind}}{\sim} \text{Bin}(45, \theta_i),$$

where H_i is the number of hits in the first 45 at-bats for player i .

- a) Using a numerical discretization approach, implement Bayesian inference in this model and obtain posterior samples of $\theta_1, \dots, \theta_N$. Then compute the Bayesian estimates (posterior means) for each θ_i . Compute the accuracy of these Bayesian estimates in two metrics: sum of squares and sum of absolute values. Compare the accuracies with the naive estimates $H_i/45$, as well as the accuracy of the James-Stein based estimates computed in class.
- b) Using the posterior samples, compute 95% credible intervals for each θ_i . Compare these intervals with the actual values of θ_i . How many of the actual θ_i 's lie in the intervals?

Player Name	Number of Hits in the first 45 At-Bats	EoS Average
Clemente	18	.346
F Robinson	17	.298
F Howard	16	.276
Johnstone	15	.222
Berry	14	.273
Spencer	14	.270
Kessinger	13	.263
L Alvarado	12	.210
Santo	11	.269
Swoboda	11	.230
Unser	10	.264
Williams	10	.256
Scott	10	.303
Petrocelli	10	.264
E Rodriguez	10	.226
Campaneris	9	.286
Munson	8	.316
Alvis	7	.200

Table 1: Some statistics of 18 baseball players from the 1970 season.

4. In class, we used the following fact:

$$\theta \sim N(\mu, \tau^2), \quad y \mid \theta \sim N(\theta, \sigma^2) \implies \theta \mid y \sim N\left(\frac{y/\sigma^2 + \mu/\tau^2}{1/\sigma^2 + 1/\tau^2}, \frac{1}{1/\sigma^2 + 1/\tau^2}\right) \quad (2)$$

and $y \sim N(\mu, \sigma^2 + \tau^2)$.

The goal of this problem is to prove a more general form of this result.

Suppose X and Y are random vectors with

$$X \sim N(m_0, Q_0), \quad Y \mid X = x \sim N(Bx, R).$$

Let

$$X \mid Y = y \sim N(m_1, Q_1).$$

a) Show that

$$m_1 = (I + Q_0 B^T R^{-1} B)^{-1} m_0 + Q_1 B^T R^{-1} y$$

and

$$Q_1 = (I + Q_0 B^T R^{-1} B)^{-1} Q_0.$$

b) Show that

$$m_1 = m_0 + Q_0 B^T (B Q_0 B^T + R)^{-1} (y - B m_0)$$

and

$$Q_1 = Q_0 - Q_0 B^T (B Q_0 B^T + R)^{-1} B Q_0.$$

c) Show that

$$Y \sim N(B m_0, B Q_0 B^T + R).$$

d) Deduce (2) as a consequence of the results in parts (a)–(c).

References

- [1] Efron, B. (2010). *Large-scale inference: empirical Bayes methods for estimation, testing, and prediction*. Cambridge University Press.
- [2] Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. (2013). *Bayesian Data Analysis*. Third edition. Chapman & Hall/CRC.